SSW 590 — DevOps

DevOpsStartup: A Startup DevSecOps Assignment Manual

Version 1.18

Student Edition

by

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Stevens Institute of Technology

September 30, 2026

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This release covers Gitea as the startup's source-control and issue-tracking hub, automated PDF builds of the team's LaTeX manual using Gitea Actions, and Jenkins continuous integration with a required merge gate. Each assignment has a 100-point rubric and requires an incremental update to the team's versioned DevOps IT manual. Further assignments can be added as the course develops.

Document History

Add the newest approved change at the top. Retain earlier entries and link each change to the affected chapter or controlled artifact.

Table 1: Assignment Manual Update History

Date / Version	**Author**	**Change**	**Changed At**
September 30, 2026 / 1.18	CDX; DDM review pending	Removed the unrelated version-control product comparison from the Gitea Actions assignment.	Chapter 3
September 30, 2026 / 1.17	CDX; DDM review pending	Explained HTTPS Git submission through the TLS proxy, internal Gitea and optional SSH ports, and the separate Actions artifact and course-PDF paths.	Chapter 3; references
September 30, 2026 / 1.16	CDX; DDM review pending	Changed copyright attribution on the copyright page and page footers to D. Darian Muresan with email address.	Front matter; page footers
September 30, 2026 / 1.15	CDX; DDM review pending	Clarified where the LaTeX toolchain runs, how Gitea Actions reaches a temporary build container, and the host-mode alternative; added a deployment diagram and matching PDF evidence requirements.	Chapter 3; Figure 3.2; references

Assignment Manual Update History (continued)

Date / Version	Author	Change	Changed At
September 30, 2026 / 1.14	CDX; DDM review pending	Added a chapter-specific block diagram of the Gitea Actions manual-build and PDF-artifact flow.	Chapter 3; Figure 3.1
September 30, 2026 / 1.13	CDX; DDM review pending	Added Gitea Actions LaTeX PDF build assignment before Jenkins, with commit/run versioning, artifact evidence, and a 100-point rubric.	Chapter 3; Chapter 1; references
September 24, 2026 / 1.12	CDX; DDM review pending	Added an optional Caddy HTTPS implementation route and official setup links without supplying a finished configuration.	Chapter 2; references
September 24, 2026 / 1.11	CDX; DDM review pending	Applied the approved `dos` namespace; aligned chapter-specific TikZ sources and included AI guidance; added a Gitea deployment reference.	Chapter 2; Chapter 4; Appendices B–C
September 23, 2026 / 1.10	DDM, CDX	Added a linked Jenkins–Gitea source citation inside the deployment diagram.	Chapter 4; TikZ figure source
September 23, 2026 / 1.9	DDM, CDX	Moved the Jenkins deployment TikZ figure into its chapter-specific source folder.	Chapter 4; TikZ figure source
September 23, 2026 / 1.8	DDM, CDX	Added a Jenkins/Gitea UML deployment diagram and required teams to adapt it in their IT manuals.	Chapter 4

Assignment Manual Update History (continued)

Date / Version	Author	Change	Changed At
September 23, 2026 / 1.7	DDM, CDX	Added the Jenkins/Gitea integration assignment, explicit PDF evidence, security guidance, and aligned 100-point rubric.	Chapter 1; Chapter 4
September 23, 2026 / 1.6	DDM, CDX	Changed the title page label to Student Edition.	Title page
September 23, 2026 / 1.5	DDM, CDX	Named Let's Encrypt in the Gitea goal and clarified that it is a reasonable free TLS option with renewal verification.	Chapter 2
September 23, 2026 / 1.4	DDM, CDX	Made the PDF-only submission and eight rubric-aligned report updates explicit; removed the template attribution page.	Chapter 1; Chapter 2
September 23, 2026 / 1.3	DDM, CDX	Limited this release to the Gitea assignment; expanded firewall and database guidance; raised optional backup/restore credit to 20 points.	Chapter 1; Chapter 2
September 23, 2026 / 1.2	DDM, CDX	Made Gitea backup and restore optional extra credit, with evidence of an isolated restore test.	Chapter 2
September 23, 2026 / 1.1	DDM, CDX	Clarified that teams submit only the compiled IT manual PDF, not its LaTeX source.	Chapter 1; future chapters retained in source
September 23, 2026 / 1.0	DDM, CDX	Created modular startup DevSecOps assignments, shared standards, student-manual update requirements, and 100-point rubrics.	Chapters 1–2; future chapters retained in source

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Chapter 1

Introduction and Common Requirements

1.1 Purpose and Chapter Map

Your team is operating a small startup, not assembling unrelated demos. Each assignment adds a working capability to the same product, repository, service, and operational manual. The assignments are modular: the instructor will announce the order and due dates, and may add, remove, or revise chapters. Each chapter states its own objective, deliverables, and 100-point rubric. This release contains the Gitea foundation, automated LaTeX manual build, and Jenkins integration assignments; later assignment chapters will be activated when they are ready.

Task	Goal
2	Establish the startup's Gitea work hub, student domain, HTTPS, and minimal public exposure.
3	Build the team's LaTeX manual on each Gitea push, embed its commit/run identity, and publish the PDF.
4	Connect Jenkins to Gitea so automated tests and a required status check protect reviewed merges.

1.2 Rules for Every Assignment

1. Maintain the team's *separate* LaTeX DevOps IT manual, beginning with the course `main/DevOpsTemplate/`. Submit only its compiled PDF on every assignment; retain the LaTeX source within the team. The manual is a living operational record, not a pasted collection of screenshots.

2. Put a version number on every submitted document. Add an update-history entry for initial setup and each assignment change, with date, author, short summary, and links to affected chapters or evidence. Explicitly name any Hosts or Passwords table changes in

that entry.

3. Keep Hosts and Passwords tables current. The Passwords table contains *hints only*: no passwords, derivation keys, recovery codes, tokens, or private keys. Use distinct secrets for distinct services and store real credentials outside the report and repository.

4. Use clear headings, labeled figures/tables, a glossary, an index, and citations where useful. Provide sufficient configuration and test evidence for another team to reproduce and verify the work. Redact sensitive values from screenshots and logs. Attribute reused code, images, and AI assistance.

5. All user-facing web interfaces must use valid hostname-based HTTPS. HTTP may be open only for redirection or certificate validation. Do not expose raw application, database, metrics, or Jenkins agent ports publicly. Restrict SSH to approved source IPs or a private access path. Document cloud firewall *and* host/container networking rules; Docker-published ports can bypass some host firewall expectations, so test exposure externally [6, 7]. Internal service-to-service traffic may use an isolated private network; document the trust boundary.

6. Use least privilege, pinned or recorded software versions, persistent data, and repeatable configuration. Never claim an unrun test passed. Describe failed attempts and corrective actions. Run disruptive experiments only on your own authorized resources, with a recovery path.

7. Record the domain offer, cloud services used, possible charges/renewals, and shutdown or cleanup steps. Student offers and prices can change; verify current terms rather than assuming they are free indefinitely.

1.3 Submission and Grading

Submit *one artifact only*: the updated, compiled PDF of the team's DevOps IT manual. Do not submit the LaTeX source, separate screenshots, configuration files, or backup archives. Put the required explanations, sanitized configuration excerpts, labeled evidence, and clickable links to the team's Gitea repository and HTTPS hostname *inside the PDF*. Keep the versioned non-secret deployment files in the team repository and link to them from the PDF. The instructor's learning-management system sets the due date and submission location. Each chapter's rubric totals 100 points. Missing evidence receives no credit for the claim it was meant to support. Security requirements are graded in the relevant rubric row; a secret exposed in a submission must be rotated and reported promptly. A missing or stale Hosts or Passwords table incurs a 10-point deduction from the assignment score (minimum zero); do not deduct for the same omission twice.

Chapter 2

Startup Foundation with Gitea

2.1 Description and Goal

Create the startup's first operational work hub on DigitalOcean. Use an existing, version-pinned Gitea container image; building an image from source is *not* required. A real startup team should be able to file an issue, review a proposed change, and find its history in Gitea. This assignment establishes the domain, HTTPS, access controls, and persistent data on which later tasks depend [15]. Let's Encrypt provides free, automated TLS certificates and is a reasonable choice for this student deployment; the team must still configure renewal and verify the live HTTPS service [28].

2.2 Required Work

1. Claim an eligible domain offer through the GitHub Student Developer Pack. Verify eligibility, first-year cost, renewal cost, and expiration [19]. Point a hostname such as `git.example.com` at the team's DigitalOcean Droplet; show the DNS record and an independent lookup.

2. Deploy Gitea using a pinned official image and versioned Compose file. Choose and document SQLite or PostgreSQL as described below. Explain the volumes, restart behavior, image source, and non-secret configuration. Prove repositories and issues survive a container replacement.

3. Put a reverse proxy or equivalent TLS endpoint before Gitea. Research two certificate options and justify one; Let's Encrypt is a reasonable option. Show a valid certificate for the hostname, HTTP-to-HTTPS redirect, expiration date, automated renewal configuration, and a successful renewal test or dry run. State where TLS terminates and how private traffic reaches Gitea. Certificate challenge choices have different port/DNS requirements [27].

4. Configure the DigitalOcean firewall and host/container publishing using the firewall requirements below. Verify the permitted and blocked ports from outside the Droplet. Keep database and application-backend ports private [6, 7].

5. Create a startup organization, product repository, README, and at least two member accounts with appropriate roles. Create at least five actionable issues, including product and operations work. Demonstrate two issue-linked branches/pull requests, peer review, merge, and issue closure.

2.3 Suggested HTTPS Implementation: Caddy

Caddy is one approachable way to satisfy the reverse-proxy requirement; it is a suggestion, not a required product. Its official Docker Compose guide shows how to run the proxy container and persist certificate data [2]. The Caddy reverse-proxy guide explains how to route a public hostname to an application, while its automatic HTTPS guide explains certificate issuance, renewal, and HTTP-to-HTTPS redirects [3, 1].

If your team chooses Caddy, use those guides to develop its own Compose and proxy configuration. Point the chosen DNS hostname at the Droplet; place Caddy and Gitea on a shared private Compose network; publish only the proxy's web ports; and route requests to Gitea by its Compose service name, not `localhost`. Persist Caddy's certificate state and set Gitea's public `ROOT_URL` to the HTTPS hostname [8]. Decide whether port 80 is needed for certificate validation and redirect, and record that decision in the firewall rules. Then verify the live certificate, redirect, renewal mechanism, and that Gitea's backend port is not publicly reachable. These are implementation goals, not a supplied configuration; the team's PDF must explain and prove its own choices.

2.4 Database: Choose, Configure, and Verify

Gitea stores issues, users, organizations, pull-request metadata, and settings in a database; Git repositories and some attachments also live in persistent application data. Both parts must survive replacement of the Gitea container. Choose *one* of the following paths and identify it in the architecture diagram [15, 9]. Figure 2.1 shows both supported alternatives; a team deploys only its selected path.

SQLite (simpler)
: Use Gitea's built-in SQLite database for a small student team. Record the database-file location inside the persistent Gitea volume (normally under `/data`), and show that the volume is mounted outside the disposable container layer. No database service or public database port is needed. Explain that SQLite is a reasonable small-team choice but may not fit a larger, higher-concurrency deployment.

PostgreSQL
: Run a version-pinned PostgreSQL container with its own persistent volume. Connect Gitea to it by Compose service name and internal port 5432 on a private Compose network; do not publish 5432 to the Droplet. Record database name, application account, image tag, volume, network, and connection settings. Put the database credential in an untracked, restricted secret file or

equivalent secret store; show a redacted configuration and explain who can read the secret. Do not include credentials in the PDF, screenshots, Compose file, or Git repository.

For either path, create a repository and an issue, record their identifiers, replace only the Gitea container without deleting volumes, and verify both remain available. Record the exact command, before/after evidence, and a short explanation of why the data persisted. Show how the database is connected to Gitea and why it is not reachable from the public internet. The persistence check is *not* a backup or restore test. Teams using PostgreSQL should also show the database container is healthy after restart and explain how Gitea behaves if that service is unavailable.

2.5 Firewall and Network Exposure

HTTPS protects traffic to the web interface; it does not close other ports. Use the DigitalOcean Cloud Firewall as the public network boundary, then check the Droplet and Docker port mappings. Apply the firewall to the correct Droplet and document its name, rules, source ranges, and date. DigitalOcean firewalls are stateful and admit only traffic allowed by their rules [6, 5]. Do not rely on host `ufw` alone: Docker-published container ports can bypass expected `ufw` filtering [7].

Service	**Public source**	**Required treatment**
443/TCP	All users	Allow to the HTTPS endpoint for Gitea.
80/TCP	All users, if used	Allow only for HTTP-to-HTTPS redirect or certificate validation; explain its purpose.
Admin SSH	Approved team IPs or private access path	Restrict port 22 (or the chosen admin port); establish and test this rule before removing broader access to avoid lockout.
Gitea backend	None	Connect proxy to Gitea privately (often port 3000); do not publish this port publicly.
PostgreSQL	None	Internal container network only (port 5432); no public host-port mapping. SQLite needs no network port.
Git over SSH (optional)	Documented users	Prefer Git over HTTPS for this assignment. If enabled, document a distinct port and the justified firewall rule; do not confuse it with admin SSH.

Document the inbound rule table for both IPv4 and IPv6 if the Droplet has both. Explain outbound rules at a high level: the Droplet must still reach needed package/image sources, DNS, time services, and certificate services. Do not blindly block all outbound traffic. Check for overlapping Cloud Firewalls whose rules might change the effective policy [6].

Provide a small network diagram showing browser to 443, optional 80 to the proxy, proxy to private Gitea backend, and Gitea to its private database or SQLite file. Figure 2.1 is a reference only: adapt it to the team's actual services, network, and database choice. Include

a sanitized Cloud Firewall screenshot or rule export, the Compose port mappings, and results of an *external* check from a machine or network outside the Droplet. Show that 443 works, 80 behaves as documented, admin SSH is restricted, and the backend/database ports cannot be reached publicly. A scan run on the Droplet itself does not prove public exposure. Record the test source, time, expected result, actual result, and any correction. Test only your own assigned Droplet.

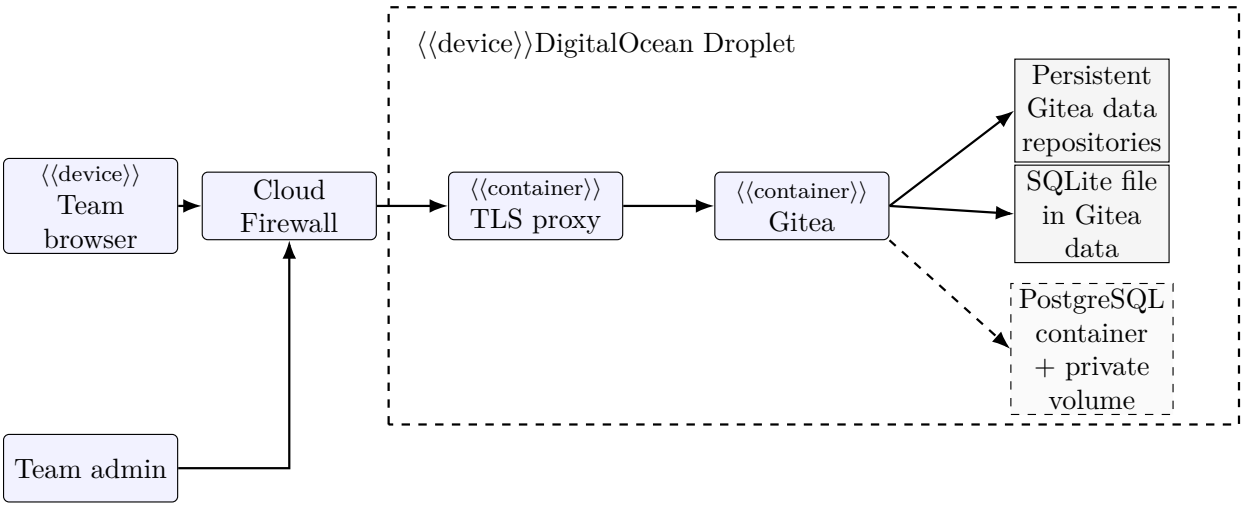


Figure 2.1: Reference Gitea deployment on one Droplet. The Cloud Firewall admits HTTPS on 443 and, when needed, HTTP on 80 for redirect or certificate validation; it limits administrator SSH. The reverse proxy alone publishes web ports. Gitea's backend and the selected SQLite or PostgreSQL data path remain private (PostgreSQL uses internal port 5432). Adapt this reference to the actual deployment [15, 6].

2.6 Exactly What to Update in the IT Manual PDF

Update the team's existing manual, then submit *only its compiled PDF*. Add a *Gitea* chapter and put the following eight numbered subsections in it. Each item matches one row of the 100-point rubric, in the same order. Embed labeled, dated, sanitized screenshots or command output where requested; put clickable links and redacted configuration excerpts in the PDF rather than submitting separate evidence files. A screenshot alone does not replace an explanation of the configuration or test result.

1. **Student Pack domain and DNS (10 points).** Identify the domain offer, eligibility, first-year and renewal costs, expiration, and the chosen Gitea hostname. Show the DNS record, independent lookup result, and HTTPS hostname link. Explain how a future team will track renewal.

2. **Pinned Gitea deployment (10 points).** Record Droplet and Gitea image versions, image source, deployment commands, restart policy, persistent mounts, and proxy connection. Include a redacted Compose excerpt and a link to the versioned non-secret

deployment files in the team repository. Explain how another team could repeat the installation.

3. **Database and persistence (10 points).** State whether the team uses SQLite or PostgreSQL and document the corresponding settings, private connection or file path, volumes, and secret handling described in Section 2.4. Give the repository/issue identifiers, exact container-replacement command, before/after evidence, and conclusion. For PostgreSQL, include the post-restart health check.

4. **HTTPS and certificate renewal (15 points).** Compare two certificate options, justify the selected one, and identify TLS termination. Show the valid hostname certificate, issuer and expiry, HTTP-to-HTTPS behavior, renewal automation, and successful renewal test or dry run. Include the relevant redacted proxy configuration and test output.

5. **Firewall and external exposure (15 points).** Include the DigitalOcean Cloud Firewall name and Droplet attachment, inbound rules and source ranges, IPv4/IPv6 treatment, host/Compose port mappings, and a small network diagram. Record external test source and time, expected versus actual results for 443, optional 80, admin SSH, Gitea backend, and database port, plus any corrections. Explain why HTTPS does not replace a firewall.

6. **Startup team workflow (20 points).** Link the organization and product repository; show the README, two members and their roles, and at least five actionable issues. Trace two issues through linked branches/pull requests, peer review, merge, and closure, using links or labeled evidence.

7. **Architecture and operation (5 points).** Explain the end-to-end request and data paths with an architecture diagram. State where data lives, which interfaces are public or private, and how the team restarts and troubleshoots the service. Distinguish the database-persistence test from a backup/restore test.

8. **Manual quality and controls (15 points).** Increase the manual's version; add a dated history entry for initial setup and this update with links to changed sections and evidence. Update the Hosts and Passwords tables (password *hints only*) and mention both in the history. Use clear headings, captions, citations, glossary/index entries where useful, and redact all secrets. State any failed tests and corrective actions.

The optional backup-and-restore evidence belongs in a separate, clearly labeled extra-credit subsection of the *same PDF*; it is not part of the eight required rubric items.

2.7 Rubric (100 Points)

Criterion	Points
Student Pack domain, cost/renewal check, DNS evidence	10
Pinned Gitea deployment and reproducible configuration	10
Database configuration and persistence verification	10
Valid HTTPS, redirect, certificate details, renewal test	15
Firewall, limited exposure, external verification	15
Startup organization, backlog, issue-to-reviewed-merge workflow	20
Architecture and operational explanation	5
Versioned IT manual, history, Hosts/Passwords hints, citations and evidence	15
Total	**100**

2.8 Optional Extra Credit: Verified Backup and Restore (Up to 20 Points)

Backup is *not* required for this assignment and earns no points in the 100-point rubric. For up to 20 extra-credit points, back up both Gitea's persistent application data (including repositories and attachments) and its database, then restore them to a separate, temporary Gitea instance. Do not overwrite the working startup service.

Document the backup method, what data it covers, where it is stored, and when it ran (5 points). Show a sanitized backup log or file listing and explain how the database and application data belong to the same recoverable state (5 points). Create a small test repository and issue before the backup, then show that both are present and usable in the restored instance (10 points). Include the restore log and labeled screenshots of the original and restored repository and issue, plus the test date and outcome. A backup file or successful backup command *alone* does not prove recovery. Do not submit backup archives, database dumps, credentials, or private keys.

Chapter 3

Automated LaTeX Manual Builds with Gitea Actions

3.1 Description and Goal

Turn the startup's DevOps IT manual into a reproducible build. Keep its LaTeX source in the Gitea repository established in Chapter 2. Each push to the chosen branch must trigger a *Gitea Actions* workflow that compiles the manual and makes the resulting PDF available as a downloadable build artifact. This assignment uses a self-managed Gitea runner, not a GitHub-hosted runner or Overleaf. Jenkins integration remains a separate assignment in Chapter 4 [11, 10].

The PDF must identify both the triggering Git commit and the workflow run. A push may contain several commits. Use the triggering commit's short hash as the source revision and the Gitea workflow run number as the build identifier. Keep the manual's human-maintained version (for example, `v1.3`) and append a build suffix such as `+build.42.a1b2c3d`. A re-run may retain the same run number; if distinguishing attempts matters, include the run-attempt number as well [12].

Figure 3.1 summarizes the expected path. It is a conceptual block diagram; teams choose and document their own runner and LaTeX toolchain.

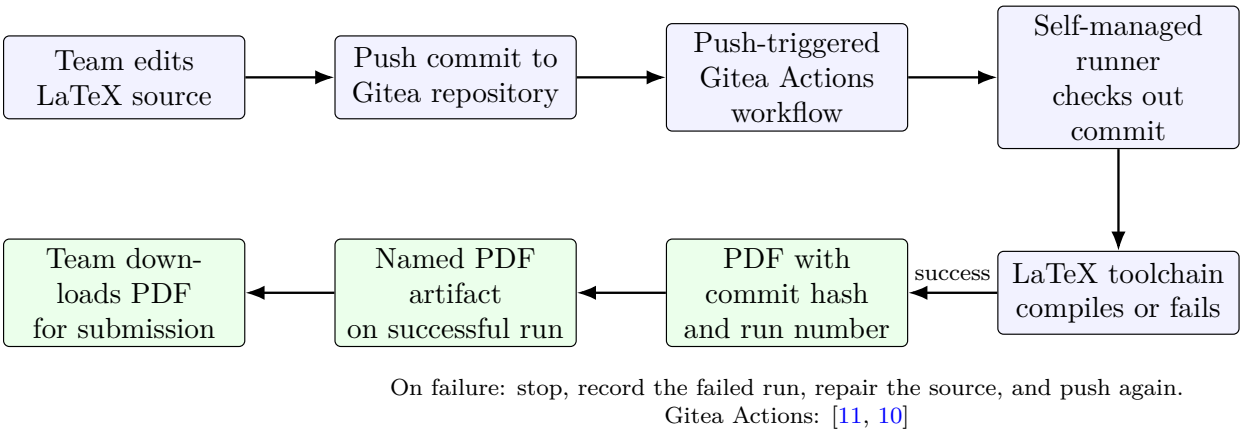


Figure 3.1: From a Gitea source push to a versioned, downloadable manual PDF.

3.2 How the Team Submits Source to Gitea

Team members commit the LaTeX project locally and use a normal `git push` to the repository's HTTPS remote, for example `https://git.example.com/team/manual.git`. The client connects to the public TLS proxy on **TCP port 443**, just as a browser does. The proxy forwards that request on the private container network to Gitea's web service, commonly on **port 3000**. Port 3000 should not be published to the internet merely to allow Git pushes. Set Gitea's public root URL to the HTTPS hostname so its repository links and clone URLs are correct [17, 15].

SSH-based Git pushes are optional and use a *separate* SSH listener: the container's SSH port is commonly 22, while the Droplet may publish it on another port if its own SSH service already uses 22. SSH does not travel through the HTTPS proxy. For this assignment, HTTPS Git pushes are sufficient; teams need not expose a Gitea SSH port. The runner has no public submission port. After a push, Gitea starts the workflow; the runner checks out that revision and uploads the finished PDF to the matching Actions run. The team downloads that PDF artifact and submits *only the PDF* for course grading, as required below [11, 10].

3.3 Where the LaTeX Compiler Runs

Recommended student architecture: keep Gitea as the Git and Actions service, run a separately registered Gitea runner, and let that runner start a temporary *job container* with a pinned TeX Live toolchain. The compiler belongs in the job environment, *not* in the Gitea application container. Gitea does not directly execute LaTeX. A runner label using Gitea's `docker://` execution mode selects the job image; the workflow's `runs-on` value selects that label. A purpose-built image avoids reinstalling TeX on every push [13, 14].

For this course manual, use **TeX Live with `latexmk -pdf`** as the starting point: `latexmk` coordinates repeat passes and `pdflatex` produces the PDF. Include the packages,

fonts, bibliography, glossary, and index tools actually used by the team's project. Record the exact image version or digest and the build command; test it against the manual before relying on Actions. A team needing different font or Unicode support may justify LuaLaTeX instead [29, 4].

The runner registers with and requests work from the Gitea HTTPS URL. For a containerized job, its Docker daemon creates the short-lived TeX job container. The job checks out the triggering Git revision from Gitea over the network, compiles the PDF inside that container, and uploads the PDF back to the workflow run as an artifact. Thus Gitea and TeX exchange source and artifacts through Actions and HTTPS; they need no shared filesystem and no additional public compiler port. Both the runner and the job container must be able to resolve and reach the Gitea URL used for the workflow [14, 10].

A runner and job containers may share the Gitea Droplet if it has enough memory and disk, or the runner may live on a separate machine. Mounting the Droplet's Docker socket into a runner is convenient but gives jobs powerful access to that host and its other containers; restrict who can edit workflows and prefer a separate runner host for stronger isolation. Installing TeX Live directly on a host-mode runner is a workable small project alternative, but jobs then use that runner environment directly and builds are less isolated. If a containerized runner has a `host` label, "host" means *inside the runner container*, not automatically on the Droplet. Installing TeX inside the Gitea container does not configure a runner to use it [13, 14].

Figure 3.2 depicts the recommended single-Droplet student option. Treat the TeX job container as temporary; the Docker daemon and runner socket are internal infrastructure, not internet-facing services.

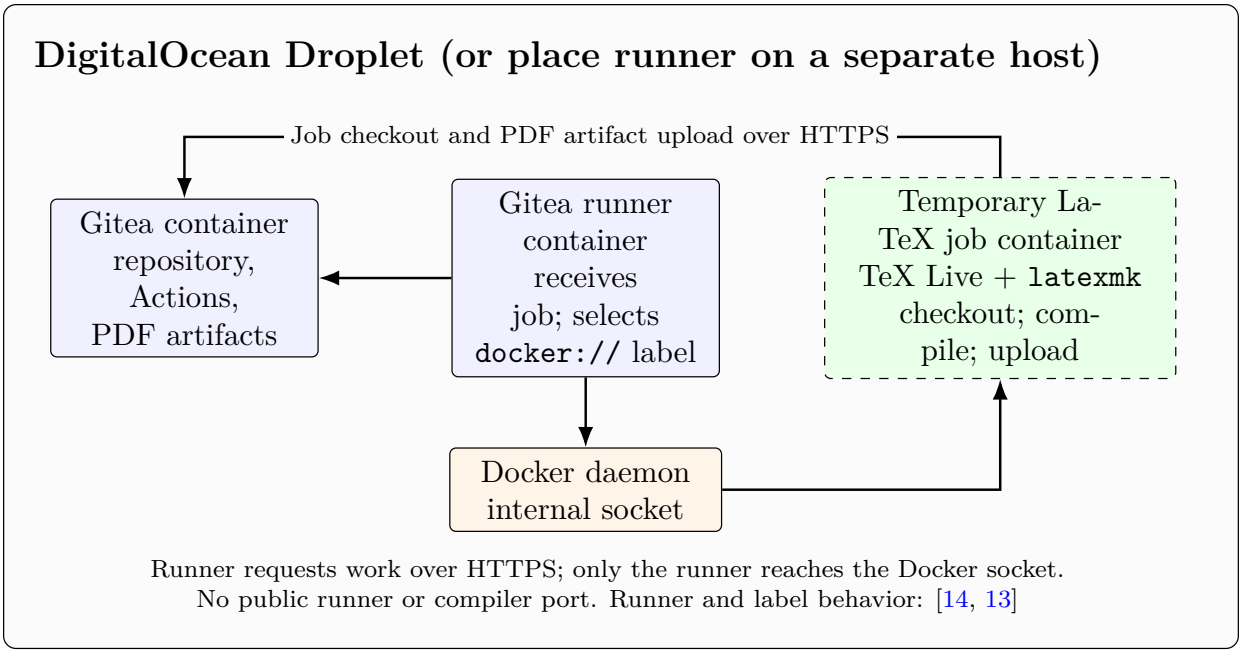


Figure 3.2: Gitea Actions runner launching an isolated LaTeX build job on the same Droplet.

3.4 Required Work

1. Put the complete, buildable LaTeX project, its bibliography, figures, and non-secret build configuration in the team's Gitea repository. Identify the main `.tex` file and a reproducible compile command. The repository is the source of truth for this assignment; editing a PDF does not satisfy the requirement. Push the source over the Gitea HTTPS remote and identify the public and internal ports in the resulting network path.

2. Enable Actions for the repository and register a Gitea runner with a label that the workflow can target. Record its location, version, execution mode, capacity, and registration scope. Ensure the runner and its job environment can reach the Gitea instance. Prefer isolated job execution; do not expose a runner management or Docker daemon port to the internet. Consider available Droplet memory and disk before placing the runner beside Gitea [11, 14].

3. Commit a workflow under `.gitea/workflows/` that runs on a push to the chosen branch. It must check out the triggering revision, prepare a recorded LaTeX toolchain and dependencies, compile the main document, and fail clearly if compilation fails or the expected PDF is absent. Do not download an unreviewed script and execute it with broad privileges. State where the compiler runs and how the job obtains its image or installed tools; do not assume the Gitea container supplies them.

4. Inject the triggering commit hash and workflow run number into the *generated PDF's* visible version information. A small generated LaTeX metadata file or command-line definitions are acceptable. Do not have the workflow commit its generated version back to the same branch: that can create a push/build loop. Keep the manual's normal version and update-history entry under the team's control [12].

5. Upload the successfully compiled PDF as a named Gitea Actions artifact. The build should not silently succeed when the PDF is missing. State how a team member finds and downloads the artifact from the matching run, and record the artifact retention setting if one is configured [10].

6. Demonstrate at least two successive source changes and pushes. For each, correlate the commit, run, visible PDF version, and artifact. Make one controlled LaTeX error on a branch covered by the workflow, then repair it and show a passing run. Do not disrupt the deployed Gitea service.

7. Protect the runner registration token and any repository secrets; do not put them in source, logs, or the PDF. Explain who may edit the workflow and why untrusted workflow changes must not receive privileged Docker access or secrets. Record the runner's access path and any Hosts/Passwords table changes using hints only.

3.5 Exactly What to Update in the IT Manual PDF

Add an *Automated Manual Build* chapter to the team's existing DevOps IT manual. The chapter must have the following eight numbered subsections, in the rubric order. Put dated, labeled, sanitized evidence and clickable Gitea links *inside the PDF*. The workflow and LaTeX source stay in Gitea; submit only the compiled PDF, not those files separately. The final submitted PDF should be downloaded from a successful workflow run.

1. **Source and trigger (15 points).** Link the repository, main `.tex` file, and workflow. Identify the selected branch and push event; show that the workflow runs after a relevant push. State the Git remote protocol, public port, TLS proxy, and private Gitea port; explain why the runner needs no public submission port.

2. **Runner setup (15 points).** Give the runner version, location, scope, label, execution mode, required network path, and capacity. Show a registered/online runner and a job assigned to it. Describe its isolation and absence of unnecessary public ports. Identify whether jobs run in temporary containers or directly in the runner environment.

3. **Reproducible compilation (20 points).** Record the toolchain or build image version, dependencies, build command, main file, and expected PDF path. Link a successful run log and explain how the build detects a compiler error or missing PDF. State where TeX is installed and how the job accesses it; explain checkout and artifact upload without assuming a shared filesystem with Gitea.

4. **PDF version traceability (15 points).** Show the manual's human-maintained version, the generated build identifier, the visible commit hash and run number in the PDF, and the matching Gitea run. Explain how metadata is injected without making an automatic commit or push loop.

5. **Downloadable artifact (10 points).** Link the named PDF artifact on the successful run and show that its downloaded file opens. State the retention setting or observed default; do not treat an artifact as a permanent backup.

6. **Change, failure, and repair evidence (10 points).** Give two successive source commits and matching runs/PDF identifiers, plus one safe failed compile and its repaired passing run. State expected and actual results; match commit IDs across evidence.

7. **Runner and secret boundary (5 points).** Describe workflow edit rights, token/secret storage without values, runner isolation, and any Docker privilege risk. Link the relevant non-secret configuration.

8. **Manual quality and controls (10 points).** Update the manual's version and dated history table; link this chapter and explicitly state whether the Hosts and Passwords tables changed. Keep those tables current, using password hints only. Include relevant citations, glossary/index terms, sanitized evidence, and a short failed-build troubleshooting procedure.

3.6 Rubric (100 Points)

Criterion	Points
Versioned source and push-triggered workflow	15
Registered runner and documented execution environment	15
Reproducible LaTeX compilation and failure detection	20
Visible commit/run version traceability in the PDF	15
Downloadable, verified PDF artifact	10
Two builds plus controlled failure and repair	10
Runner permissions and secret boundary	5
Versioned IT manual, history, tables, and evidence	10
Total	**100**

Chapter 4

Continuous Integration with Jenkins

4.1 Description and Goal

Add automated verification to the same startup product repository established in Chapter 2. A Gitea pull request should trigger a Jenkins pipeline, display a status on the proposed commit, and be unmergeable while the required check fails. After a repair, the team should be able to review and merge it. Use an existing Jenkins container image; building one from source is not required. A small, real test suite is preferable to a long pipeline that proves little. Jenkins and its plugins must be configured as a controlled service, not an unauthenticated public endpoint [24, 23, 26].

4.2 Required Work

1. Deploy an official, recorded Jenkins LTS image using a versioned, non-secret Compose configuration. Persist `/var/jenkins_home`; record the image and installed plugin versions, restart policy, and available Droplet memory/storage. Create named administrator accounts and verify that Jenkins settings and jobs survive a container restart. A backup/restore test is *not* required for this assignment [24].

2. Give Jenkins its own hostname, such as `ci.example.com`, with a valid HTTPS certificate. The existing reverse proxy and a service such as Let's Encrypt may be reused. Set the Jenkins URL and reverse-proxy headers correctly; show certificate validity, redirect behavior, and renewal configuration. Require login and appropriate permissions. Document the DigitalOcean Cloud Firewall, host firewall, and Compose port mappings. Expose only the needed HTTPS entry point; do not publish raw Jenkins port 8080 or an agent port to the internet. Restrict browser access to approved team source ranges or a private access path, in addition to Jenkins login [25, 26].

3. Integrate the startup's Gitea repository with Jenkins. The Jenkins Gitea plugin and Multibranch Pipeline are a supported route, but another documented, working integration is acceptable. Configure a repository-scoped webhook for relevant push and pull-request events, or explain and demonstrate a polling fallback. If both services are on the same

Droplet, use a private container network for webhook delivery where practical; do not open the entire Jenkins interface merely to receive a webhook. Record the event, delivery result, triggering commit, and matching Jenkins run. Use a dedicated Gitea integration identity and only the permissions actually needed for repository access and status reporting [18, 23].

4. Commit a `Jenkinsfile` to the product repository. It must check out the proposed commit, run at least one meaningful build or lint step and one automated test, and report an unambiguous pass or failure. Use the product's actual language/framework. Record where dependencies come from, how a clean run is reproduced, and which stage failed. A command that merely prints "success" is not an adequate test.

5. Protect the default branch in Gitea and require the Jenkins status check on the pull request head commit before merge. Identify the exact check context; do not confuse a successful webhook delivery or a green Jenkins UI page with a passing merge gate. Show that a failed check blocks merging and that a passing check permits the team's normal peer-review process. The Gitea Checks plugin is one way to publish status, but a proven equivalent is acceptable [16, 22].

6. Demonstrate a controlled failure and repair on one issue-linked pull request: show the failed commit, failed test/run, blocked merge, corrective commit, passing run/status, peer review, and eventual merge. Use real links and matching commit identifiers so the sequence can be audited. Do not intentionally break the deployed Gitea or Jenkins services.

7. Keep tokens and passwords in Jenkins credentials or an approved secret store, never in the repository, Compose file, console log, or report. Keep credential scope narrow and explain who can change the `Jenkinsfile`. Pipeline code executes commands: prefer a separate build agent with zero executors on the controller. For a small student deployment that runs builds on the controller, explicitly document the risk, restrict pipeline authors to trusted team members, and do not grant untrusted pull requests access to secrets or privileged Docker access. Record a plugin-update check and the operator response to a failed webhook or build [21, 20].

4.3 Network and Trust Boundary

Show browser-to-proxy-to-Jenkins and Gitea-to-webhook paths separately. A private webhook path does not make the browser interface private, and HTTPS does not replace firewall rules or authentication. If Gitea and Jenkins are on different Droplets, use a documented private network or a narrowly reachable HTTPS webhook endpoint rather than exposing port 8080. Explain which source addresses can reach each path. Check public exposure from outside the Droplet, including IPv6 if enabled. Before narrowing SSH or browser access, verify an approved administrator can still connect. Do not publish an inbound agent port solely for this assignment [18, 25].

Figure 4.1 is a reference deployment, not a prescribed implementation. It shows the same-Droplet option; teams using separate Droplets must draw the actual network boundary and webhook route. Replace the Gitea data node with the SQLite volume or PostgreSQL service and volume chosen in Chapter 2. Show a separate build agent if one is deployed.

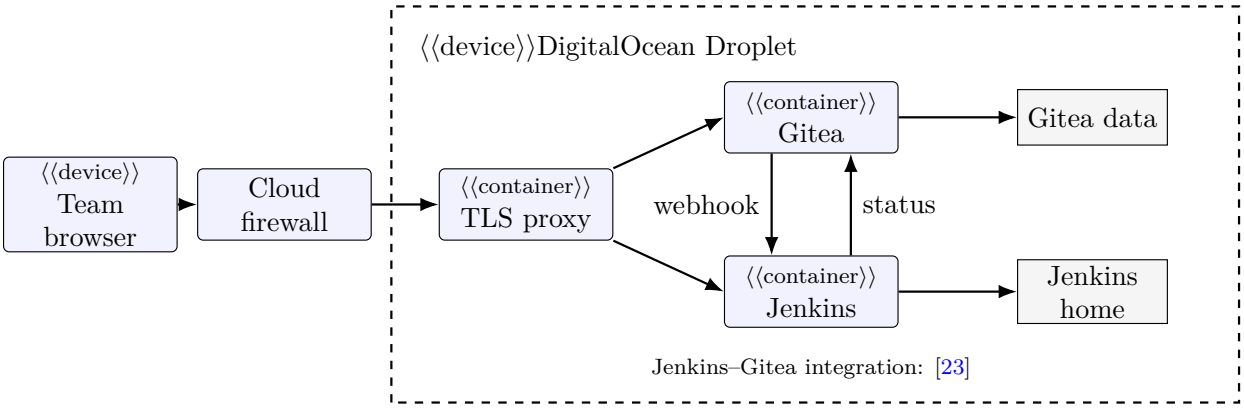


Figure 4.1: UML deployment reference for one-Droplet Gitea and Jenkins integration. The browser reaches the TLS proxy through firewall-approved HTTPS (443). Webhook, repository checkout, status, and data paths stay private. The Gitea data node represents the team's chosen database and persistent repository storage.

4.4 Exactly What to Update in the IT Manual PDF

Submit *only* the updated, compiled PDF of the team's existing LaTeX DevOps IT manual. Add a *Jenkins/CI* chapter with the following eight numbered subsections, in the same order as the rubric. Include labeled, dated, sanitized evidence and clickable links *inside the PDF*; keep the versioned `Jenkinsfile` and non-secret deployment configuration in Gitea and link to them. Do not submit those files separately. Explain each test's expected and actual result; screenshots alone are insufficient.

1. **Jenkins deployment and persistence (10 points).** Record the official image source and tag, plugin versions, Compose link and redacted excerpt, Droplet capacity, persistent volume, restart policy, and container restart test. Show the Jenkins configuration and job still present afterward. State explicitly that restart persistence is not a backup test.

2. **HTTPS, access, and firewall (15 points).** Link the Jenkins hostname. Show DNS, certificate issuer/expiry, HTTP-to-HTTPS behavior, renewal configuration, Jenkins URL/proxy settings, named accounts and roles, Cloud Firewall rules, and host/Compose port mappings. Give dated external test results for HTTPS, raw port 8080, and any agent port; include IPv6 where applicable and identify the approved browser-access path.

3. **Gitea trigger and traceability (15 points).** Adapt Figure 4.1 as a labeled UML deployment diagram of your actual browser and webhook paths. Identify the Gitea repository, integration identity, event selection, webhook/polling configuration, delivery result,

commit ID, and linked Jenkins run. Explain one failed delivery diagnosis or the steps the operator would take to diagnose one.

4. **Versioned pipeline and tests (15 points).** Link the `Jenkinsfile`; describe checkout, build/lint, and automated-test stages, dependencies, expected outputs, and clean-run instructions. Include a sanitized run view with distinct stage results and the tested commit ID.

5. **Required merge gate (15 points).** Show the default-branch protection setting, exact required status-check context, failed check blocking a pull-request merge, and passing check allowing the usual review and merge. Explain how the status reaches Gitea from Jenkins.

6. **Failure, repair, and review (10 points).** Give an issue and pull-request link and a dated sequence of failed commit/test, blocked merge, repair commit, passing run/status, peer review, and merge. Match commit IDs across Gitea and Jenkins.

7. **Credentials and build boundary (10 points).** Identify secret storage and scope without revealing values; state who can edit the `Jenkinsfile`, where builds execute, how untrusted contributions are kept from secrets, and the plugin-update check. Include a short webhook/build troubleshooting procedure. If builds run on the controller, document the specific risk and compensating restrictions.

8. **Manual quality and controls (10 points).** Increase the PDF's version; add a dated history entry linking this chapter and naming updates to the Hosts and Passwords tables. Update both tables (password *hints only*), the UML deployment diagram, relevant citations, glossary/index entries, and an operator restart/troubleshooting note. Redact secrets and report any failed tests and corrective actions.

4.5 Rubric (100 Points)

Criterion	Points
Jenkins deployment and persistence	10
HTTPS, access control, and firewall verification	15
Gitea trigger and traceable Jenkins run	15
Versioned pipeline with meaningful build and tests	15
Required Gitea status check and merge gate	15
Failed-then-repaired, reviewed pull request	10
Credentials, pipeline trust boundary, and plugin operations	10
Versioned IT manual, history, Hosts/Passwords hints, and evidence	10
Total	**100**

Appendix A

Student IT Manual Update Guide

The assignment manual tells teams what to do; the team's own DevOps IT manual records what they actually built, tested, and learned. Start from `main/DevOpsTemplate/` and preserve its versioning, history, glossary, index, citation, and consistent chapter conventions. Add one operational chapter per capability as it is delivered. A new chapter should explain purpose, architecture, configuration, verification, security, failure modes, recovery, ownership, and cost/cleanup. Link figures, commands, repository files, and evidence with stable labels. Do not copy an assignment brief as if it were an implementation report.

A.1 Minimum Table Fields

Table	**Suggested fields**
Hosts	Hostname, role, environment, provider, public/private address or DNS, exposed ports, owner, HTTPS URL, backup/monitoring status.
Passwords	Service/account, owner, approved vault reference or non-revealing hint, privilege level, MFA/rotation status. Never the secret itself.
History	Date, version, author, what changed, why, and linked chapter/table or test evidence. Mention Hosts and Passwords updates explicitly.

A.2 Evidence Standard

Separate *planned*, *configured*, and *verified* work. State the test date, expected result, actual result, and evidence location. Include negative results and fixes. Screenshots should be legible and captioned, but versioned configuration and commands are usually more reproducible. Before submission, verify hyperlinks, citations, compilation, redaction, and the consistency of every table and architecture diagram.

Appendix B

AI Agent Usage Guide

This LaTeX chapter is the authoritative AI-maintenance directive for the assignment manual. An AI agent should read it, the introduction, the affected assignment chapter, and the current history before editing. Make changes only to the requested scope. The instructor-approved file namespace is `dos`. Project-owned filenames begin with `dos`, followed by a PascalCase responsibility and file extension (for example, `dosStartupFoundation.tex`). The root source is `dosManual.tex`. `prologue.tex`, `cornell.cls`, and `Makefile` are retained exceptions for front matter, an externally maintained class, and a conventional build entry point; generated build files are not authored sources.

Keep one responsibility per chapter. A new assignment chapter needs a description, observable deliverables, explicit student-manual PDF evidence, and a rubric totaling exactly 100 points. Keep future assignment chapters commented out until the instructor activates them. Place each TikZ source at `TikZ/<including-file-base>/<dosDrawingName>.tex`; the directory must exactly match the base name of the chapter containing its `\input`. The TikZ file contains only the `tikzpicture`; the owning chapter contains the figure environment, caption, label, and citations. Keep substantive AI instructions in this LaTeX chapter and log material AI-assisted work in Appendix C; any tool-discovery file should point here rather than duplicate these rules.

Do not invent due dates, budgets, eligibility, tested outcomes, security posture, cloud access, grades, or instructor approval. Treat the LaTeX source as authoritative and the PDF as a review artifact. Never include secrets, tokens, private keys, student personal data, or unredacted production logs. Verify current external product documentation before adding instructions that depend on versions or offers. Update the document version and history for content changes. Build the complete PDF, inspect the log and affected pages, and record material AI assistance in Appendix C. The instructor decides assignment scope, deadlines, exceptions, and grading policy.

Appendix C

AI Assistance Log

Record material AI-assisted edits so later instructors can distinguish human-approved requirements from generated draft text. Do not paste private student data or credentials into prompts or this log.

Date	**Tool / Reviewer**	**Task and human verification**
2026-09-24	OpenAI Codex / instructor review pending	Applied the instructor-approved `dos` namespace, aligned the TikZ source layout and AI chapters with the architecture template, and added a Chapter 2 Gitea deployment reference. Built and checked the PDF; the instructor should confirm the diagram fits the intended student deployment choices.
2026-09-23	OpenAI Codex / D. Muresan	Drafted modular startup DevSecOps assignment chapters and rubrics from the instructor's stated course goals; the instructor should review scope, order, deadlines, and rubric policy before release. LaTeX build/visual checks recorded in the working session.

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Glossary

DevSecOps A DevOps practice that integrates security throughout planning, building, delivery, and operations. 23

evidence A retained, sanitized record of an actual execution, observation, or review. 23

least privilege Giving an account or service only the access it needs for its assigned task. 23

reverse proxy A service that receives external requests and forwards them to internal services; it may terminate HTTPS. 23

rollback A controlled return to a known acceptable version or state. 23

runbook A documented, repeatable operational procedure. 23

Acronyms

AI artificial intelligence. 23

CD continuous delivery. 23

CI continuous integration. 23

MFA multi-factor authentication. 23

SLO service-level objective. 23

TLS Transport Layer Security. 23

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